Ritwick Verma

Search and Retrieval Infrastructure · C++ · Systems
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SUMMARY

Backend engineer with 5.5 years on Cassini, the distributed C++ search engine behind every ebay.com search (1B+ listings). Owned vehicle-parts fitment and built retrieval, caching and query-transform features, shipped through A/B tests. 215 merged PRs, in the top 10% of 121 contributors to the engine.

EXPERIENCE

eBay

Software Engineer 3, Search Backend, San Francisco, CA — Jun 2023 to Present

- Owned eBay Motors fitment, the C++ subsystem that decides which parts fit a buyer's vehicle, across 16 sites and 19 catalogs. Designed the query operator clients use to ask for fit constraints or a per-item "fits" signal.
- Built a request-scoped result cache that removed repeat backend calls across recall-recovery passes. Requests that already carry embeddings now skip the embedding service, large requests are batched under its limits Also fixed an embedding cache that had stopped hitting.
- Designed three C++ query transforms for the product-grouped carousels on the results page, checked against replayed production traffic. Folded three ranking expressions into one native C++ macro that matched production exactly.
- Made the tokenizer stack allocator-aware (C++ PMR) in 7 PRs and shipped it as an HTTP API for other teams.
- Replayed 50,000 production-sampled queries and caught an initialization-order bug before release, and the fix cut new errors by about 96%. Also hotfixed a production parser crash and capped a combinatorial blow-up in query rewriting.
- Wrote the markdown-based test frameworks the team adopted as its standard, a language server for the engine's query language (719 operators), and a CLI that diffs query results across environments.

Software Engineer 2, Search Backend, San Jose, CA — Mar 2021 to Jun 2023

- Rewrote fitment for garage vehicles and tire and wheel specifications (88 merged PRs in 2022), with spec translation for the German, French and Italian sites, and added A/B testing to it.
- Reworked the engine's core query, table and property containers to use C++ polymorphic allocators.

Earlier: software intern at Shoelace Wireless (Android, 2020) and BHEL (Java EE, 2017).

SELECTED PROJECTS

Side projects are written with AI coding agents. I made the design calls and checked the results. Also ShutUp, an Android app with 26,000+ downloads (v1 written by hand in 2016).

- **Self-hosted agent:** An LLM agent on my homelab with tools for code, mail, documents and infrastructure. It runs the homelab and does most of the typing on the projects below.
- **TizenAmbilight:** Ambilight from inside a Samsung TV. The public capture API only returns the UI layer, so it reads the undocumented video plane instead, tone-maps HDR and drives an LED strip at 20 to 25 fps. The image decoders were checked byte for byte against libjpeg-turbo.
- **Crush 80 firmware:** Open ZMK firmware for a keyboard whose vendor never released source: a Telink RISC-V chip, 154 per-key LEDs on a ported SPI-DMA driver, and a pinout recovered from PCB photos and the stock binary.
- **Homelab as code:** 16 Proxmox guests behind Traefik, Authelia SSO and a Headscale VPN. 14 are declared in OpenTofu, a nightly CI job commits config drift, and alerts fire once per real fault.
- **Plutus:** Python research setup for trading strategies with point-in-time backtests and block-bootstrap confidence intervals. Five popular strategies failed the tests and were dropped.

SKILLS

- **Languages:** C++ (17/20/23), Python, Java, Kotlin, C#, C
- **C++ tooling:** Polymorphic allocators (PMR), ASan, clang-tidy, CMake, GoogleTest
- **Search:** Query rewriting, tokenization, recall and ranking factors, embedding retrieval, caching, A/B tests
- **AI tools:** Claude Code, agent tooling and MCP servers
- **Systems:** Linux, Docker, Proxmox, OpenTofu, GitLab CI, Traefik and OIDC

EDUCATION

Master of Computer Science, University of California, Irvine — 2019 to 2020
B.Tech, Computer Science and Engineering, SRM Institute of Science and Technology — 2015 to 2019